ISO
14001:2015

COURSE SLIDES · WSQ

ISO 14001:2015 Internal Auditor Training

WSQ Course Code: TGS-2025054479 · 2-Day Instructor-Led Training

Skills Framework TSC: Internal Audit Engagement Planning (ACC-IAD-3005-1.1)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN: 201200696W

Version v5 · 17 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR with your phone camera
3. Submit your attendance — AM, PM and Assessment
4. Repeat on BOTH days of the course
5. 75% minimum attendance for assessment and funding

Digital attendance is mandatory for WSQ-funded courses and is the record SSG uses to determine your funding eligibility.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

BACKGROUND
PhD — 20+ years of industry and training experience in management systems, quality and data analytics.

DELIVERS
WSQ courses on ISO management system auditing, quality, sustainability and data science.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your experience with ISO 14001, EMS or internal auditing (if any) — and whether your organisation is certified.
- One environmental issue at your workplace you would like to be able to audit confidently by the end of Day 2.

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HOUSEKEEPING

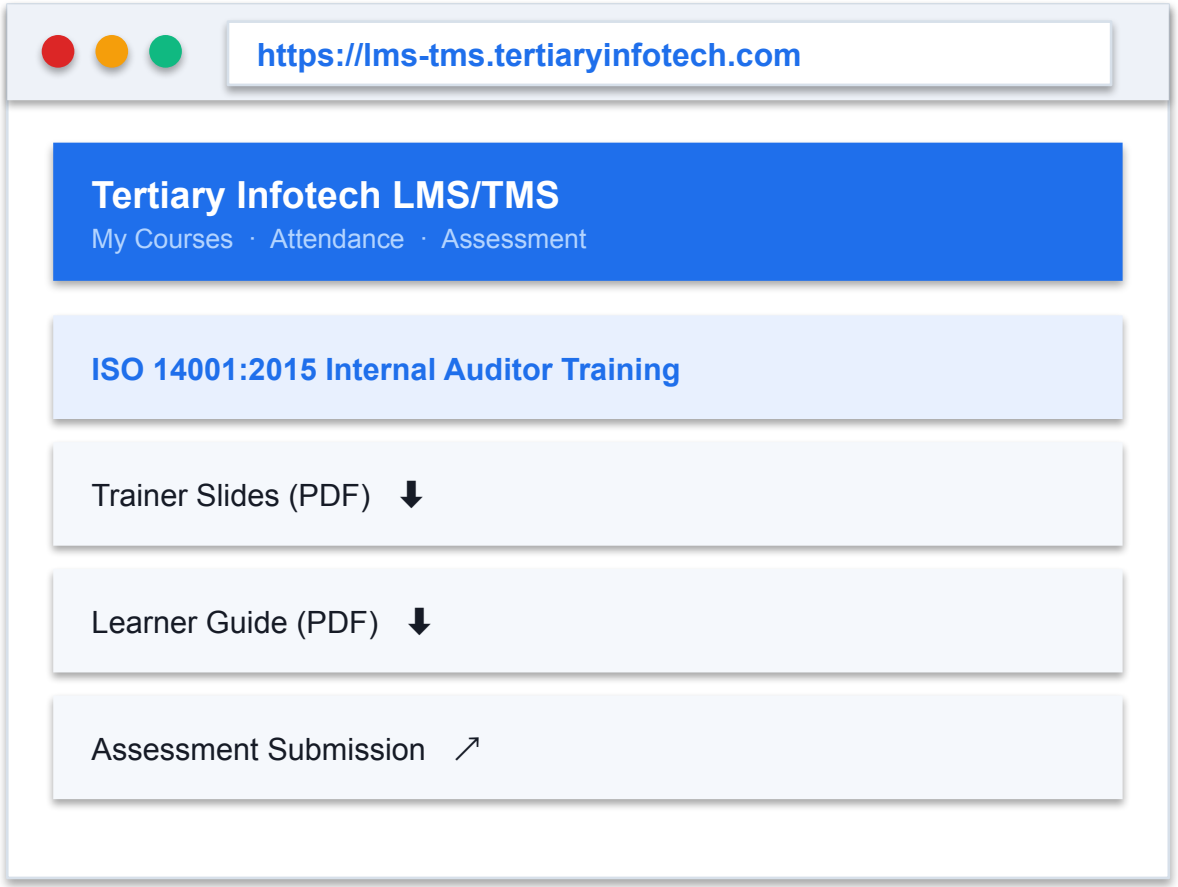
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding.

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COURSE PORTAL · LMS/TMS

Download Course Material



1 **Sign in**
lms-tms.tertiaryinfotech.com — log in with your registered email.

2 **Open your course**
Select this course under My Courses.

3 **Download materials**
Slides, Learner Guide and the activity briefs — your open-book references.

4 **Submit & attendance**
Submit your assessment answers and scan the digital attendance QR here.

All course material is downloaded from the LMS/TMS portal — keep it handy: both assessments are open book.

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WSQ ALIGNMENT

Skills Framework Alignment

TSC element	Detail
TSC Title	Internal Audit Engagement Planning
TSC Code	ACC-IAD-3005-1.1 · Proficiency Level 1
Knowledge K1	International Professional Practices Framework (IPPF) comprising definitions of Internal Auditing, Code of Ethics, core principles and International Standards for the Professional Practice of Internal Auditing
Knowledge K2	Organisation's significant risks, objectives, resources, operations, businesses and structure
Knowledge K3	Internal audit methodologies and procedures of the organisation
Knowledge K4	Risk identification and assessment techniques
Ability A1	Identify relevant frameworks, guidelines and standards to be applied in the engagement
Ability A2	Identify stages of the engagement and various stakeholders involved
Ability A3	Support conduct of preliminary survey and risk assessment

Every topic, activity and assessment item in this course maps to these TSC knowledge and ability statements.

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BY THE END OF THIS COURSE

Learning Outcomes

1

LO1

Identify ISO 14001:2015 standards for internal auditing based on the International Professional Practices Framework (IPPF) and organisational risks

2

LO2

Identify phases of engagement with stakeholders by applying ISO 14001:2015 internal audit methodologies and procedures

3

LO3

Support preliminary risk assessments using risk identification and assessment techniques

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THE JOURNEY

Course Outline — Three Topics Over Two Days

DAY 1 — the standard and the framework

- **Topic 1 — ISO 14001:2015 Internal Auditor Framework**
  - Overview of the IPPF
  - Introduction to ISO 14001:2015 and the EMS
  - Applying ISO 14001:2015 to the organisation — risks, objectives, resources, operations
  - Activity 1 — scoping a real engagement
- **Topic 2 begins — audit methodologies**

DAY 2 — auditing and risk assessment

- **Topic 2 — Auditing ISO 14001:2015**
  - Phases of auditing and stakeholder engagement
  - Activity 2 — audit plan, checklist and interviews
- **Topic 3 — Risk Assessment**
  - Risk assessment · corrective action requests · audit reports
  - Activity 3 — risk, CARs and the report
  - Final assessment: WA (SAQ) then Case Study

14 hours of facilitation + 2 hours of assessment = 16 hours in total.

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SCHEDULE · DAY 1

Lesson Plan — Day 1

Time	Session	Method
9:30–9:45	Digital Attendance (AM) · Welcome and course introduction · Trainer…	Admin · Ice-breaker
9:45–11:15	LU1: Introduction to ISO 14001:2015 · T1: ISO 14001:2015 Internal Auditor…	Lecture · Discussion
11:15–11:30	Tea Break	—
11:30–13:15	T1 continued (K1, K2)	Lecture · Worked examples…
13:15–14:00	Lunch Break	—
14:00–15:30	Digital Attendance (PM) · T1 continued — Apply ISO 14001:2015 to the…	Lecture · Case study ·…
15:30–15:45	Tea Break	—
15:45–17:30	LU2: Auditing Methodologies · T2: Auditing ISO 14001:2015 (K3, A2)	Lecture · Discussion
17:30–17:45	Day 1 recap · Q&A · preview of Day 2	Review · Q&A

7 hours of facilitation (9:30 am – 5:45 pm). Digital attendance is taken in the morning and again in the afternoon.

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SCHEDULE · DAY 2

Lesson Plan — Day 2

Time	Session	Method
9:30–9:45	Digital Attendance (AM) · Day 1 recap	Admin · Review
9:45–11:15	T2 continued — Phases of Auditing (K3, A2)	Lecture · Worked examples
11:15–11:30	Tea Break	—
11:30–13:00	T2 continued — Engagement with Stakeholders (A2) · ACTIVITY 2 — AquaPure…	Case study · Role play ·…
13:00–13:45	Lunch Break	—
13:45–15:30	Digital Attendance (PM) · LU3: Risk Assessment Techniques	Lecture · Worked examples…
15:30–15:45	Tea Break	—
15:45–17:30	T3 continued — Corrective Action and Reporting (K4, A3) · ACTIVITY 3…	Case study · Group work ·…
17:30–17:45	Course summary · Briefing for Assessment · Assessment Digital Attendance ·…	Admin · Briefing
17:45–18:45	FINAL ASSESSMENT — Written Assessment, Short Answer Questions (WA/SAQ) · 4…	Assessment
18:45–19:45	FINAL ASSESSMENT — Case Study (CS) · 3 scenario-based questions covering A1…	Assessment

7 hours of facilitation + 2 hours of assessment (9:30 am – 7:45 pm). Course total: 14 h facilitation + 2 h assessment.

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BEFORE THE ASSESSMENT

Briefing for Assessment

1. **Phones away**
   Place phones and all other materials under the table or on the floor.

2. **No recording**
   No photographs or recording of the assessment scripts.

3. **Work alone**
   No discussion with other learners during the assessment.

4. **Black or blue pen**
   No liquid paper and no correction tape.

5. **Raise your hand**
   The assessor will come to you if you have a question.

6. **Time is called**
   Scripts are collected when the time is up.

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Final Assessment

1

Written Assessment (SAQ)

4 short-answer questions · 60 minutes · covers K1–K4.

2

Case Study (CS)

3 scenario-based questions · 60 minutes · covers A1–A3.

3

Open book

Slides, Learner Guide, activity briefs and your own notes.

4

Individual work

Both instruments are completed individually.

5

Grading

Competent (C) / Not Yet Competent (NYC) — all answers must be correct to be Competent.

6

Appeals

An appeal process is available if you disagree with the outcome.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ) 1 hr, then the Case Study 1 hr — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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ELIGIBILITY

Criteria for WSQ Funding

1

Attendance

Minimum 75% attendance based on the SSG digital attendance record.

2

Assessment

Complete both instruments and be assessed as Competent (C).

3

TRAQOM

Complete the TRAQOM survey on the LMS.

4

SkillsFuture Credit

Eligible individuals may use SkillsFuture Credit to offset course fees. See www.skillsfuture.gov.sg.

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TOPIC 01

ISO 14001:2015 Internal Auditor Framework

IPPF · the EMS and Annex SL · Clauses 4–10 · aspects, impacts and compliance obligations · risk-based thinking

01

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WHY WE ARE HERE

An internal audit is not an inspection.

It is a systematic, independent and documented process for obtaining evidence and evaluating it objectively against agreed criteria.

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THE ROLE

What an Internal EMS Auditor Actually Does

1

Verifies conformity

Tests the EMS against ISO 14001:2015 AND the organisation's own requirements — clause 9.2.1 requires both.

2

Tests effectiveness

Not just 'is there a procedure' but 'is it implemented, maintained and working'.

3

Gathers objective evidence

Records, documented information, observations and interview responses — never opinion.

4

Reports to management

Findings go to the people who can act on them, and feed clause 9.3 management review.

5

Supports improvement

Findings drive corrective action under clause 10.2 and continual improvement under 10.3.

6

Stays independent

Auditors do not audit their own work — clause 9.2.2 requires objectivity and impartiality.

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K1 · KNOWLEDGE

The International Professional Practices Framework

The professional framework that governs how internal auditors behave and work

1.1

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FOUR MANDATORY COMPONENTS

The IPPF — Mandatory Guidance

1 Definition of Internal Auditing

An independent, objective assurance and consulting activity designed to add value and improve an organisation's operations, using a systematic, disciplined approach.

2 Code of Ethics

Integrity · Objectivity · Confidentiality · Competency — the rules of conduct that govern auditor behaviour.

3 Core Principles

The fundamental characteristics of effective internal auditing: integrity, competence, due professional care, objectivity, alignment with strategy, and insight.

4 International Standards

The Standards for the Professional Practice of Internal Auditing — attribute standards (who we are) and performance standards (what we do).

THE POINT

All four are MANDATORY. Conformance is required for anyone claiming to practise internal auditing under the IPPF — and each one is examinable in this course.

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The Code of Ethics — Four Principles in Practice

1

Integrity

Perform work with honesty, diligence and responsibility. Do not knowingly be party to any illegal activity.

2

Objectivity

Maintain an unbiased mental attitude. Do NOT audit an area where you have operational responsibility.

3

Confidentiality

Respect the value and ownership of information received; do not disclose it without authority.

4

Competency

Apply the knowledge, skills and experience needed. Decline — or get support for — work you are not competent to do.

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Independence and Objectivity — the Two Failure Modes

✗ IMPAIRED — do not do this

- **Auditing the department you work in**
  - Auditing a procedure you personally wrote
  - Auditing your own corrective actions from last year
  - Accepting gifts or hospitality from the auditee
  - Consulting mid-audit: 'what you should do is buy a new bund'
  - Being pressured to downgrade a finding

✓ INTACT — this is the standard

- **Auditing a department other than your own**
  - Declaring any prior involvement before the audit starts
  - A second auditor verifies your own area
  - Findings graded on evidence, not on how the auditee reacts
  - You note the gap; the auditee designs the fix
  - Impartiality recorded in the audit plan

Clause 9.2.2 requires the organisation to 'select auditors and conduct audits to ensure objectivity and the impartiality of the audit process'. In a small company, cross-departmental auditing is the usual solution.

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K1 · KNOWLEDGE

ISO 14001:2015 and the Environmental Management System

What the standard is, how it is structured, and what an EMS is for

1.2

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THE STANDARD IN ONE SLIDE

What is ISO 14001:2015?

1

An EMS standard

Specifies requirements for an environmental management system an organisation can use to enhance its environmental performance.

2

Certifiable

The requirements are auditable; organisations can be certified to it by an accredited certification body.

3

No performance levels

The standard does NOT state specific environmental performance criteria — the organisation sets its own.

4

No exclusions

Unlike ISO 9001, there is no clause-exclusion mechanism. All requirements must be incorporated and fulfilled.

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CLAUSE 1 · WHAT THE EMS IS FOR

The Three Intended Outcomes of an EMS

1 Enhancement of environmental performance

The EMS must actually improve how the organisation performs environmentally — not merely document what it already does.

2 Fulfilment of compliance obligations

Legal requirements the organisation has to comply with, and other requirements it has to or chooses to comply with.

3 Achievement of environmental objectives

Objectives set by the organisation, consistent with its environmental policy and its significant environmental aspects.

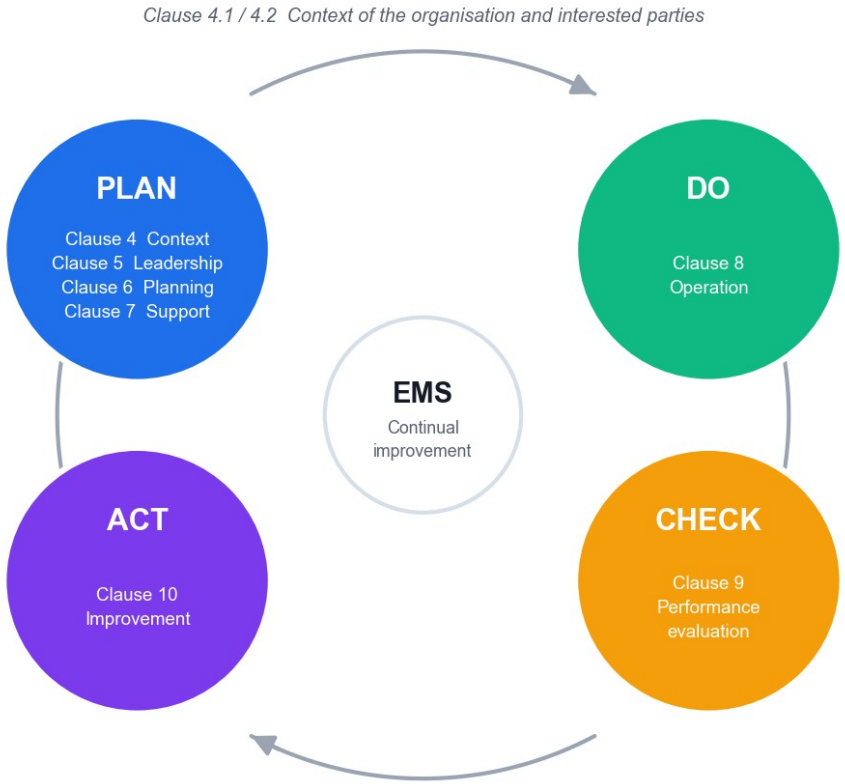
THE POINT

Every audit finding you write should ultimately connect to one of these three outcomes. If it does not, ask yourself whether it is really a finding.

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THE STRUCTURE

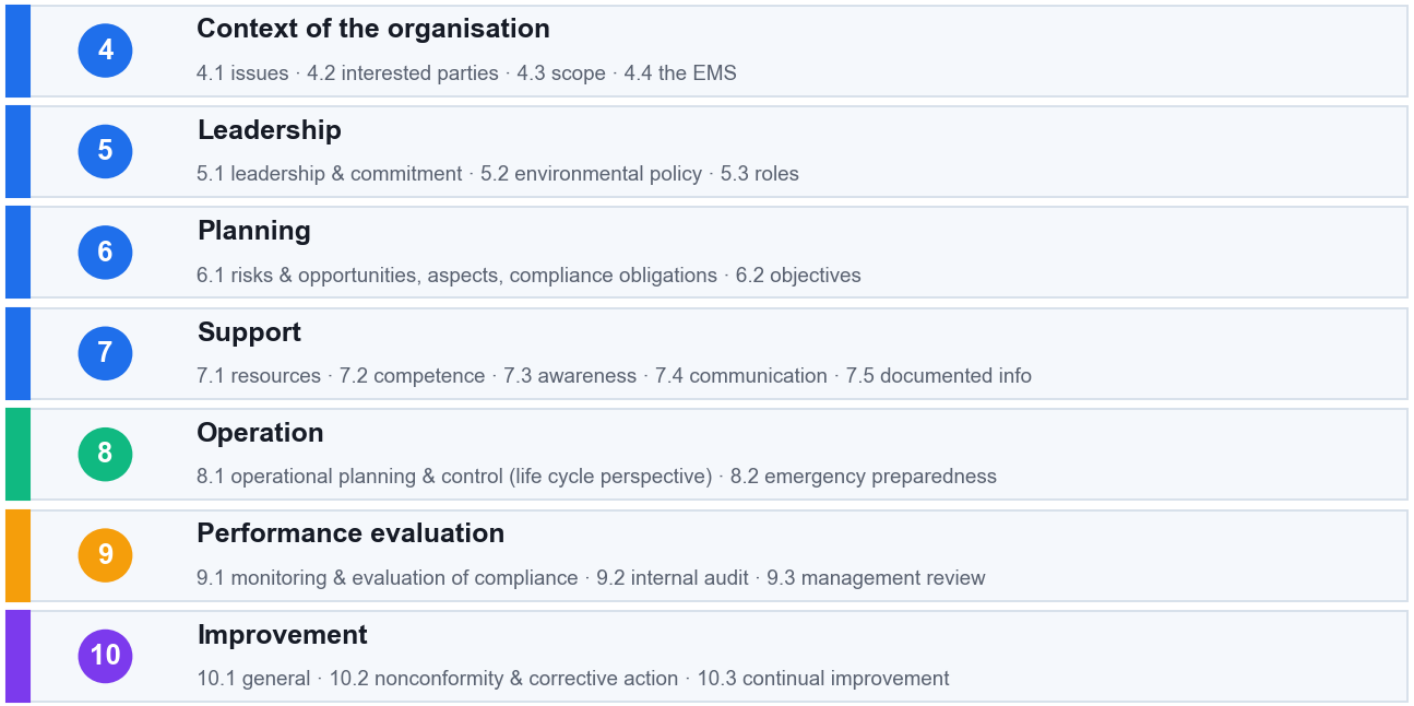
PDCA — the Engine of the Standard



Plan (clauses 4–7) · Do (clause 8) · Check (clause 9) · Act (clause 10). Context (4.1/4.2) encircles the whole cycle. Continual improvement is the purpose, not a by-product.

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The 10 Clauses — Annex SL High Level Structure



Clauses 1–3 are Scope, Normative references and Terms. Clauses 4–10 carry every auditable requirement. The identical high level structure is what lets ISO 14001 integrate with ISO 9001 and ISO 45001.

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Verbal Forms — What Is Actually Auditable

Form	Meaning	Auditable?
shall	A requirement	YES — non-fulfilment is a nonconformity
should	A recommendation	No — cannot raise an NC against a 'should'
may	A permission	No
can	A possibility or capability	No

ISO 19011 is written entirely in 'should' — it is GUIDANCE. You audit against ISO 14001 clause 9.2, never 'against ISO 19011'.

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THE 2015 REVISION

What Changed: ISO 14001:2004 → 2015

NEW OR STRENGTHENED

- **Context of the organisation (4.1, 4.2)**
  - Two-way: the organisation affects the environment AND environmental conditions affect the organisation
- **Leadership (5.1) — top management must DEMONSTRATE leadership**
- **Risk-based thinking (6.1) — risks and opportunities**
- **Life cycle perspective (6.1.2, 8.1)**
- **Environmental performance emphasis (9.1.1, 10.3)**

REMOVED OR RENAMED

- **Preventive action — DELETED**
  - The whole of clause 6.1 IS the preventive mechanism. Never raise an NC for 'no preventive action procedure'
- **Management representative — no longer mandatory**
  - Duties remain but may be distributed (5.3)
- **'Documents' + 'records' → 'documented information'**
- **'Objectives and targets' → 'environmental objectives'**

The single most common auditor error on a transitioned EMS: raising a finding for a missing preventive-action procedure or a missing environmental manual. Neither is required by ISO 14001:2015.

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Clause 4 — Context of the Organisation

1 4.1 Issues

Determine external and internal issues relevant to the EMS — INCLUDING environmental conditions capable of affecting the organisation. No documented information required.

2 4.2 Interested parties

Determine the parties, their relevant needs, and CRUCIALLY which of those needs the organisation adopts as compliance obligations.

3 4.3 Scope

Boundaries and applicability. MUST be maintained as documented information and be available to interested parties.

4 4.4 The EMS

Establish, implement, maintain and continually improve the EMS including the processes needed and their interactions.

THE POINT

Audit tip: 4.1 and 4.2 require NO documented information. Do not demand a 'context document' — test by interview and by tracing whether context actually shaped the aspects, obligations and objectives.

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Clause 5 — Leadership

1 5.1 Leadership and commitment

Top management shall DEMONSTRATE leadership: take accountability for EMS effectiveness, integrate EMS requirements into business processes, ensure resources. Non-delegable — interview top management directly.

2 5.2 Environmental policy

Commitments to protection of the environment (including prevention of pollution), to fulfil compliance obligations, and to continual improvement. Maintained as documented information; communicated; available to interested parties.

3 5.3 Roles and responsibilities

Assigned AND communicated. Specific assignment for EMS conformity and for reporting EMS performance to top management. The 'management representative' is no longer mandatory.

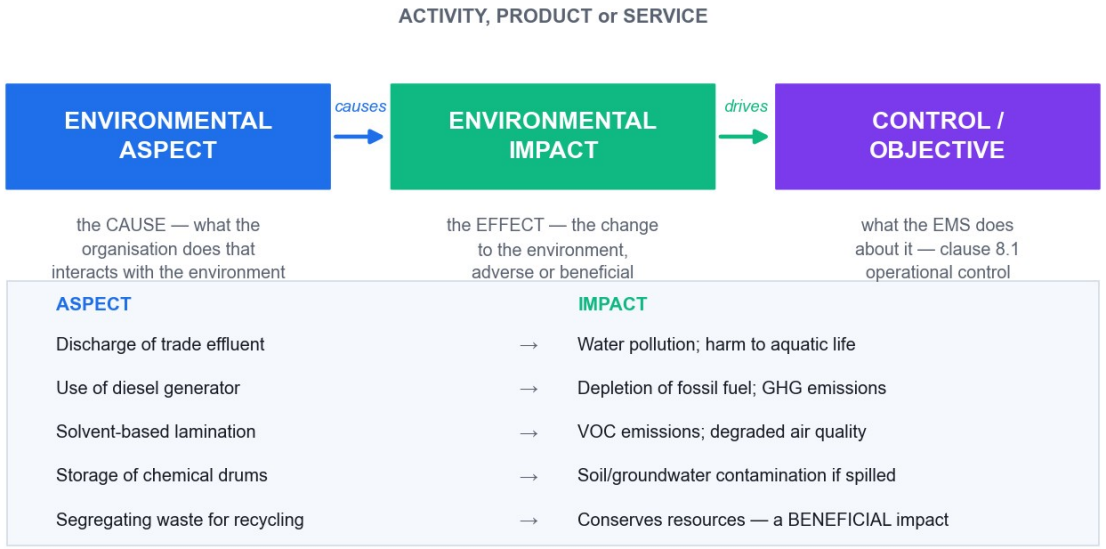
THE POINT

The strongest leadership test is not the signed policy on the wall — it is whether top management can describe the organisation's significant environmental aspects without being prompted.

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THE CORE CONCEPT OF THE STANDARD

Clause 6.1.2 — Environmental Aspects



ASPECT		IMPACT
Discharge of trade effluent	→	Water pollution; harm to aquatic life
Use of diesel generator	→	Depletion of fossil fuel; GHG emissions
Solvent-based lamination	→	VOC emissions; degraded air quality
Storage of chemical drums	→	Soil/groundwater contamination if spilled
Segregating waste for recycling	→	Conserves resources — a BENEFICIAL impact

Aspect = the CAUSE

The element of your activity that interacts with the environment. You control it.

Impact = the EFFECT

The change to the environment — adverse OR beneficial.

Control AND influence

6.1.2 covers aspects you can control and those you can influence — not just inside the fence.

Criteria decide significance

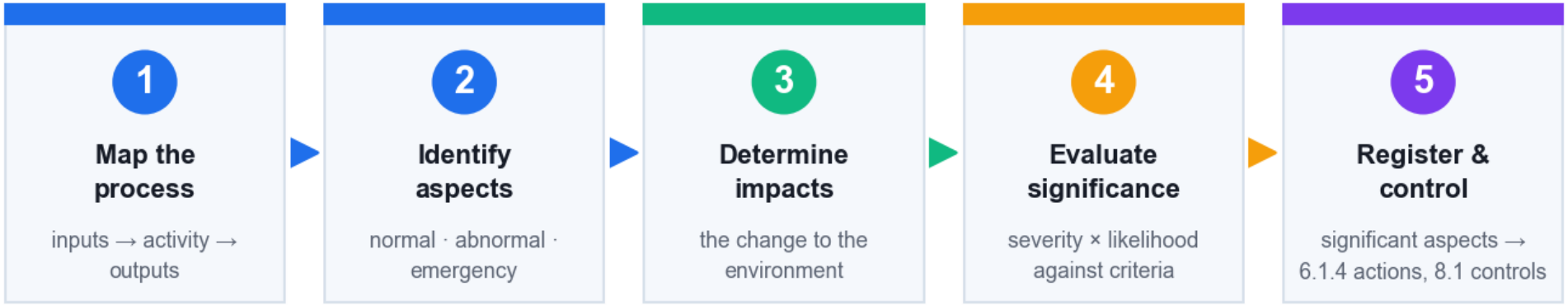
Determine significant aspects BY USING ESTABLISHED CRITERIA. The criteria are auditable.

The classic register errors: writing 'diesel generator' (that is equipment), or 'pollution' (that is an impact) in the aspect column.

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CLAUSE 6.1.2 IN PRACTICE

Determining Aspects — the Required Process



Clause 6.1.2 requires aspects to be determined for activities, products and services the organisation can CONTROL and those it can INFLUENCE — considering a life cycle perspective, and past, present and planned activities.

Clause 6.1.2 explicitly requires consideration of CHANGE (including new or modified activities), of ABNORMAL conditions and of REASONABLY FORESEEABLE EMERGENCY situations — and a life cycle perspective.

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GET THE COLUMNS RIGHT

Aspects and Impacts — Worked Examples

Activity	Environmental ASPECT (cause)	Environmental IMPACT (effect)
PCB washing	Discharge of trade effluent containing copper	Watercourse contamination; toxicity to aquatic life
Solvent lamination	Emission of VOCs from solvent-based adhesives	Air pollution; photochemical smog
Chiller operation	Consumption of grid electricity; HFC refrigerant leakage	Resource depletion; GHG emissions and climate change
Chemical drum storage	Potential release from bund failure (EMERGENCY)	Soil and groundwater contamination
Waste segregation	Recovery of recyclable material	Conserves resources — a BENEFICIAL impact

A register with zero beneficial aspects and zero emergency aspects is a red flag — it usually means the register was copied, not built.

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WHAT A GOOD REGISTER COVERS

The Three Dimensions of a Complete Aspect Register

1. **Normal**
   Routine, steady-state operation as designed.

2. **Abnormal**
   Predictable but non-routine: maintenance, cleaning, changeover, start-up and shut-down — often the WORST emissions per hour.

3. **Emergency**
   Reasonably foreseeable: spill, fire, bund failure, power loss. Links directly to clause 8.2.

4. **Past · present · planned**
   Legacy contamination, current operations, and planned changes — assessed BEFORE the change.

5. **Direct (control)**
   Own emissions, own effluent, own waste, own energy.

6. **Indirect (influence)**
   Supplier practices, contractor behaviour, product use, end-of-life. Omitting these entirely is a common finding.

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Clause 6.1.3 — Compliance Obligations

1 Legal requirements

What the organisation HAS TO comply with: EPMA and its regulations, EPHA, the Sewerage and Drainage Act, the Resource Sustainability Act, the Carbon Pricing Act, licences and permit conditions.

2 Other requirements

What it HAS TO or CHOOSES TO comply with: customer contracts, corporate policies, industry codes, agreements with community groups.

3 The three duties

(a) determine and HAVE ACCESS TO them; (b) determine HOW THEY APPLY; (c) TAKE THEM INTO ACCOUNT in the EMS. Maintained as documented information.

THE POINT

A bare register of law titles does not meet 6.1.3 — the organisation must have determined HOW each obligation applies to it. That interpretation is what you audit.

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Singapore Compliance Obligations — What to Ask For

Regime	Regulator	Evidence an auditor asks to see
Air emissions (EPMA)	NEA	Source emission test reports from an accredited lab; stack/CEMS records; calibration certificates
Trade effluent to SEWER	PUB	Written Approval and its conditions; sampling results; pre-treatment O&M logs
Trade effluent to WATERCOURSE	NEA	Discharge limits (e.g. pH 6–9, TSS 50 mg/L); accredited lab analyses
Toxic industrial waste	NEA	Collector's licence covering THAT waste; e-consignment notes (WRMS); transport approvals
Boundary noise	NEA	Monitoring by a Type 1 meter; calibration certificates; complaint register
Carbon tax	NEA	Registration; Monitoring Plan; verified Emissions Report; calibration records
Hazardous substances	NEA	Hazardous Substances Licence and permits; transaction records; SDS

Verified as at August 2026. Limits, thresholds and rates change — audit the METHOD of maintaining the legal register (6.1.3) and the evaluation of compliance (9.1.2), not just this year's numbers.

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FIGURES VERIFIED AUG 2026

Verified Singapore Figures — Use With a 'Verified As At' Date

Item	Figure	Source
Carbon tax 2026–2027	S$45 per tCO_2e	NCCS / Carbon Pricing Act
Carbon tax — taxable facility	≥ 25,000 tCO_2e per year	Carbon Pricing Act
Carbon tax — reportable facility	≥ 2,000 tCO_2e per year (report, no tax)	Carbon Pricing Act
Boundary noise — factory, day	75 dB(A) over any 5-min period	EPM (Boundary Noise) Regs
Trade effluent to watercourse	pH 6–9 · TSS 50 mg/L · COD 100 mg/L	NEA
Energy Conservation Act trigger	≥ 54 TJ/yr in 2 of 3 preceding years	ECA 2012

Teach learners to date every figure in a legal register and to re-verify at a defined frequency — that discipline is what clause 6.1.3 and 9.1.2 actually require.

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THE TWO-WAY TEST

Clause 4.1 — Context Works BOTH Ways

1

The organisation affects the environment

Emissions, effluent, waste, resource use — the traditional view, and still clause 6.1.2.

2

The environment affects the organisation

NEW in 2015: 'environmental conditions capable of affecting the organisation'.

3

External issues

Regulation, climate, resource scarcity and price, technology, market and competitive pressure, community expectations.

4

Internal issues

Activities, products and services, strategic direction, culture, capabilities, and planned change.

5

Climate change is explicit

The Feb 2024 amendment requires the organisation to determine whether climate change is a relevant issue.

6

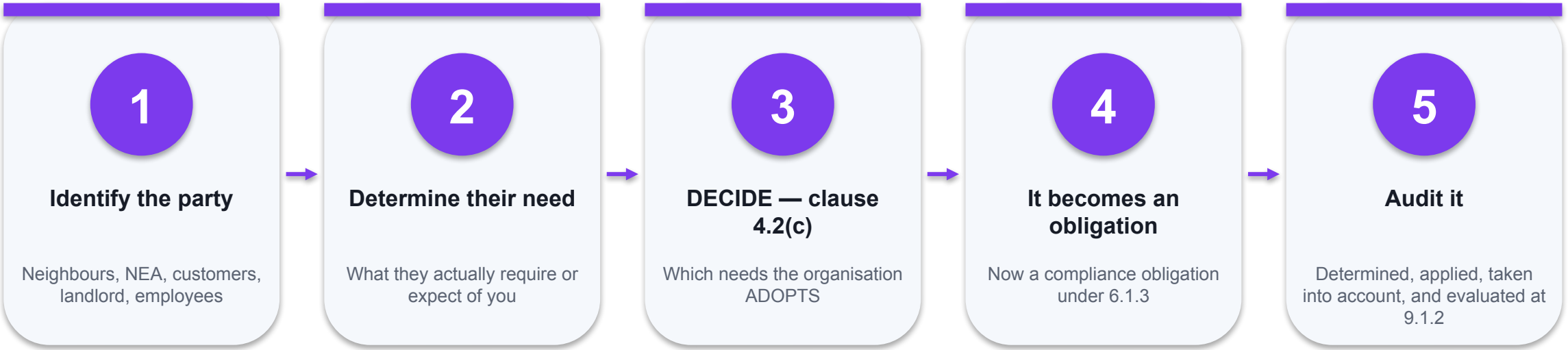
No document required

4.1 mandates no documented information — test whether context actually shaped the aspects, obligations and objectives.

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CLAUSE 4.2(C) · THE GATEWAY

How an Interested Party Need Becomes Auditable



THE POINT

Not every expectation is a compliance obligation. A customer request becomes one the moment it is written into a contract the organisation signs — or the organisation otherwise chooses to comply. That decision point is exactly what you audit.

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TAKE THESE ON SITE

Identifying Aspects — Four Methods That Actually Work

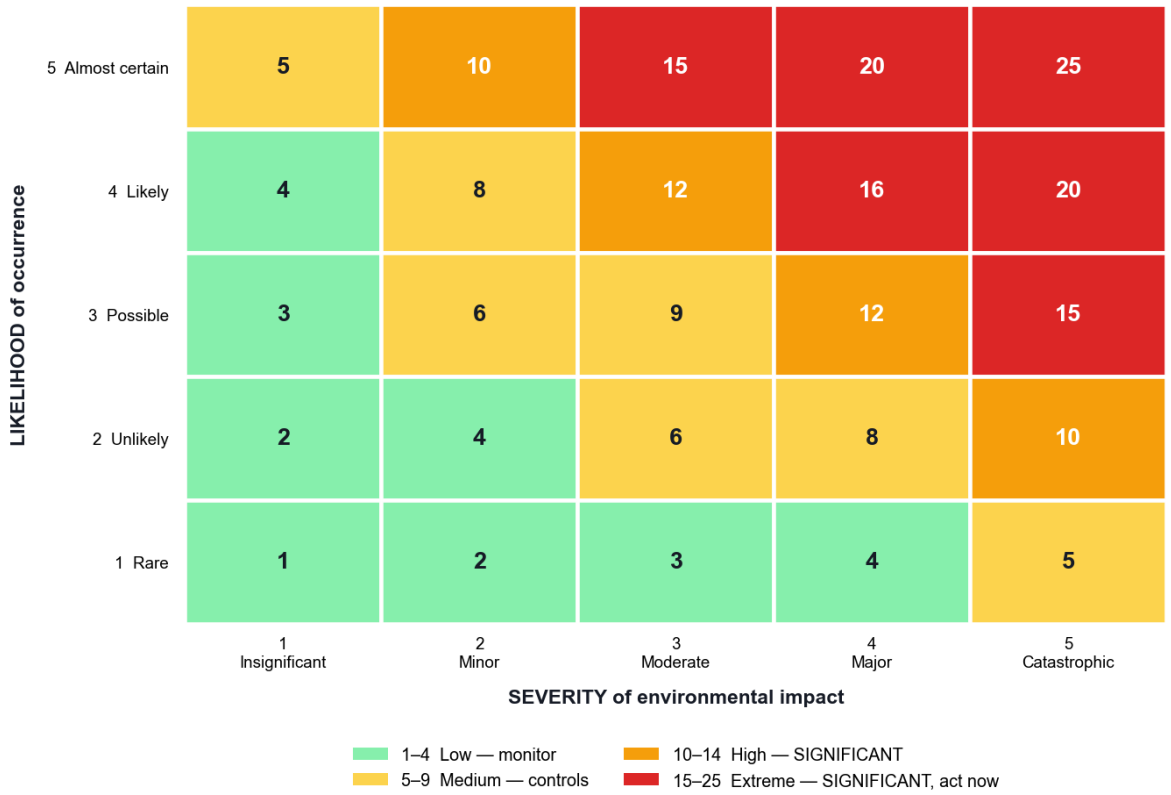
Method	How it works	What it catches
Process mapping	Flowchart each step, interrogate each for interaction	Aspects inside repeatable production processes
Input–output analysis	Materials, water, energy, chemicals IN → product, emissions, effluent, waste OUT	The MASS BALANCE gap — 'you bought 500 L of solvent and disposed of 120 L. Where did the rest go?'
Site walkthrough	Walk the drains, stacks, vents, bunds, skips, storage areas	The informal and undocumented — the unbunded drum behind the workshop
Life cycle stages	Raw materials → design → production → transport → use → end of life	Indirect aspects outside the fence that a register often misses

The mass-balance question is the single most productive aspect-audit technique: unaccounted input is almost always an unregistered fugitive emission.

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THE MOST MISUNDERSTOOD PART OF THE 2015 STANDARD

Risks and Opportunities are NOT Aspects



Aspect (6.1.2)

Your interaction with the environment. Technical and physical.

Risk / opportunity (6.1.1)

Threats and opportunities to the EMS achieving its intended outcomes. Strategic.

Aspects FEED risks

Aspects are one input to 6.1.1 — along with compliance obligations and 4.1/4.2 issues.

The common failure

Relabelling the aspect register as the 'risk register' does NOT meet clause 6.1.1.

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Clause 6.2 — Environmental Objectives

1 6.2.1 The objective

Established at relevant functions and levels, taking account of significant aspects and associated compliance obligations. Consistent with policy, measurable if practicable, monitored, communicated, updated.

2 6.2.2 The plan — five parts

WHAT will be done · WHAT RESOURCES · WHO is responsible · WHEN it completes · HOW results are evaluated, including indicators.

3 The audit test

Trace each objective BACKWARDS to a significant aspect or obligation, and FORWARDS to 9.1 monitoring evidence and 9.3 management review.

THE POINT

'Comply with all laws' and 'zero incidents' are not improvement objectives — they are baseline expectations. Challenge them.

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Clause 7 — Support

1 7.1 Resources · 7.2 Competence

Determine and provide resources. Determine necessary competence for work affecting environmental performance and the ability to fulfil compliance obligations — including contractors. RETAIN evidence of competence.

2 7.3 Awareness

Policy, the significant aspects related to THEIR work, their contribution to EMS effectiveness, and the implications of not conforming. No documented information required — test by interview.

3 7.4 Communication

What, when, with whom and how. Internal communication must enable staff to CONTRIBUTE to continual improvement. External as required by compliance obligations.

4 7.5 Documented information

Creating and updating (identification, format, review and approval); control (availability, protection, distribution, storage, change control, retention).

THE POINT

Awareness ≠ competence. Ask an operator: 'which environmental aspects relate to YOUR job, and what happens if this goes wrong?' — that is a 7.3 test, and it is where paper systems fall over.

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Clause 8 — Operation

1 8.1 Operational planning and control

Establish OPERATING CRITERIA and control processes in accordance with them. Control planned changes and review unintended changes. Ensure outsourced processes are controlled or influenced.

2 8.1 Life cycle perspective — four duties

Environmental requirements in design and development · procurement requirements · communicate requirements to external providers including contractors · consider providing information on transport, use and end-of-life.

3 8.2 Emergency preparedness and response

Prepare, respond, mitigate, PERIODICALLY TEST where practicable, review and revise (especially after an incident or a test), and provide information and training.

THE POINT

A full Life Cycle Assessment is NOT required. Ask instead: 'which life cycle stages did you consider, what did you conclude you can control or influence, and why?'

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Clause 9 — Performance Evaluation

1 9.1.1 Monitoring and measurement

What to monitor, methods ensuring valid results, the CRITERIA and INDICATORS for evaluating performance, and when. Equipment must be CALIBRATED OR VERIFIED.

2 9.1.2 Evaluation of compliance

Determine the FREQUENCY of evaluation, evaluate and act, and MAINTAIN KNOWLEDGE AND UNDERSTANDING of compliance status. Retain the results.

3 9.2 Internal audit

At planned intervals, testing conformity to ISO 14001 AND the organisation's own requirements, and effective implementation. The programme must consider environmental importance, changes, and previous results.

4 9.3 Management review

Seven required inputs and six required outputs — including implications for the STRATEGIC DIRECTION of the organisation.

THE POINT

9.1.2 is NOT the same as 6.1.3, and NOT the same as 9.2. 6.1.3 asks 'which obligations apply?'; 9.1.2 asks 'are we actually complying?'; 9.2 audits the system.

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Clause 10 — Improvement

1

10.1 General

Determine opportunities for improvement from 9.1, 9.2 and 9.3, and implement actions to achieve the intended outcomes of the EMS.

2

10.2 Nonconformity and corrective action

React and correct · evaluate the need to eliminate the CAUSE · determine whether similar nonconformities exist or could occur · implement · REVIEW EFFECTIVENESS · change the EMS if necessary. Retain evidence of both the NC and the results.

3

10.3 Continual improvement

Continually improve the suitability, adequacy and effectiveness of the EMS TO ENHANCE ENVIRONMENTAL PERFORMANCE.

THE POINT

10.2(d) — 'review the effectiveness of any corrective action taken' — is the clause that makes 'closed without verification' a nonconformity in its own right.

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15 MANDATORY ITEMS · 7 MAINTAIN + 8 RETAIN

Documented Information — MAINTAIN vs RETAIN

	MAINTAIN (a document — how to do it)	RETAIN (a record — evidence it was done)
Clause 4–6	4.3 scope · 5.2 policy · 6.1.1 risks & opportunities · 6.1.2 aspects, criteria, significant aspects · 6.1.3 obligations	6.2.1 environmental objectives
Clause 7	—	7.2 competence · 7.4.1 communications
Clause 8	8.1 operational control · 8.2 emergency	—
Clause 9	—	9.1.1 monitoring results · 9.1.2 compliance evaluation · 9.2.2 audit programme and results · 9.3 management review
Clause 10	—	10.2 nature of NCs, actions taken, and results of corrective action

There is NO required documented procedure anywhere in ISO 14001:2015, and NO environmental manual. Never raise a nonconformity for their absence.

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TRANSITION AWARENESS

ISO 14001:2015 was withdrawn on 15 April 2026 and replaced by ISO 14001:2026.

Certificates issued to the 2015 edition remain valid through a 36-month transition. The 10-clause high level structure is retained. This course teaches the 2015 edition because that is what most auditees are still certified to.

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ACTIVITY 1 · 45 MIN

Activity 1 — Auditing GreenPack — Framework, Context and…

GreenPack Manufacturing has expanded capacity by 40% and added a solvent-based thermal lamination line. Energy use is up 25%, VOC emissions have risen and three odour complaints have been received — but the aspects register and policy are stale. You are the lead internal auditor: scope the engagement.

1. Read the scenario and list the IPPF components that govern your conduct
2. Map the symptoms to ISO 14001:2015 clauses (4.1–4.4, 5.2, 6.1.2, 8.1)
3. Build a context table: issues, interested parties, compliance obligations
4. Decide the audit scope, boundaries and the areas you will prioritise
5. Write the engagement note justifying scope against risk

YOU'LL PRODUCE A completed Audit Engagement Scoping Note for GreenPack — clause map, context table, prioritised scope and the IPPF basis for your independence.

Materials: Activity 1 brief · ISO 14001:2015 clause map · Context & scope template · Full step-by-step guide: Learner Guide, Activity 1

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Activity 1 — Check Your Work

YOU ARE DONE WHEN

Your scoping note names at least 8 relevant clauses each tied to a specific scenario fact, ranks the lamination line as the top priority with a stated reason, and contains an explicit auditor-independence statement.

IF YOU ARE STUCK — WHAT USUALLY GOES WRONG

- You listed clauses but did not tie each one to a fact from the scenario — the clause map must be evidence-led, not a contents page.
- You allocated equal audit time to all three departments. Audit effort follows environmental risk and change; say so and defend it.
- You treated the odour complaints as the finding. They are a SYMPTOM — the stale aspects register is the systemic issue.
- You forgot the independence statement. The IPPF requires objectivity to be considered and recorded before the engagement starts.
- You wrote 'the policy is old' without naming clause 5.2 and what it requires.

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Tea Break

15 minutes

TOPIC 02

Auditing ISO 14001:2015

ISO 19011:2018 principles · audit programme vs audit · the six-stage lifecycle · checklists, evidence, interviewing · stakeholders

02

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GET THIS RIGHT FIRST

ISO 19011 tells you HOW to audit. ISO 14001 clause 9.2 says you MUST.

ISO 19011:2018 is written entirely in 'should' — it is guidance, not requirements. You never raise a nonconformity 'against ISO 19011'.

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The Seven Audit Principles — ISO 19011:2018 Clause 4

1. **Integrity**
   The foundation of professionalism — perform work honestly, diligently and responsibly.

2. **Fair presentation**
   The obligation to report truthfully and accurately — including unresolved disagreements.

3. **Due professional care**
   Diligence and judgement appropriate to the importance of the task and the confidence placed in the auditor.

4. **Confidentiality**
   Security of information — auditors must not use information for personal gain or to the detriment of the auditee.

5. **Independence**
   The basis for impartiality and objectivity of the audit conclusions — auditors do not audit their own work.

6. **Evidence-based approach**
   The rational method for reaching reliable and REPRODUCIBLE conclusions in a systematic process.

7. **Risk-based approach**
   An audit approach that considers risks and opportunities — new in the 2018 edition.

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THE DISTINCTION MOST COURSES SKIP

Audit Programme vs Individual Audit



Clause 5 manages the PROGRAMME — the whole schedule of audits for the year. Clause 6 conducts ONE audit. ISO 14001 clause 9.2.2 requires the programme to consider environmental importance, changes, and previous audit results.

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TWO DIFFERENT CONVERSATIONS

Programme Findings vs Audit Findings

PROGRAMME-LEVEL (ISO 14001 cl. 9.2.2)

- The audit procedure has not been updated after major change
- Equal audit time regardless of environmental risk
- Previous nonconformities closed without verification
- Auditors not competent or not impartial
- Results not reported to relevant management
- The programme itself is not monitored or improved

AUDIT-LEVEL (this one audit)

- A checklist that does not cover the audit criteria
- Evidence not verified before a finding is raised
- A nonconformity statement missing its evidence
- No opening or closing meeting held
- The report omits the sampling limitation
- Findings not agreed with the auditee before reporting

Fixing a programme finding improves every future audit. Fixing an audit finding improves one. Know which one you are looking at.

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ISO 19011:2018 §6.2 – §6.7

The Six-Stage Audit Lifecycle



THE POINT

Every commercial provider researched teaches this same sequence. Stage 6 is the one organisations most often skip — and skipping it is what turns 'closed' nonconformities into repeat findings.

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Clause 5 — Managing the Audit Programme

1. **Establish programme objectives**
   Driven by management priorities, the significant aspects, compliance obligations and previous results.

2. **Determine programme risks and opportunities**
   Resourcing, scheduling, auditor competence, access, and ineffective follow-up.

3. **Establish the programme**
   Frequency, methods, responsibilities, planning requirements and reporting — ISO 14001 cl. 9.2.2.

4. **Implement it**
   Schedule audits, select and assign competent auditors, manage records.

5. **Monitor it**
   Are audits happening on schedule? Are findings being closed? Are auditors performing?

6. **Review and improve it**
   Feed the programme's own performance back into next year's programme.

ISO 14001 cl. 9.2.2 requires the programme to take into account the environmental importance of the processes concerned, changes affecting the organisation, and the results of previous audits. All three are directly auditable.

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ISO 19011 A.6

Sampling — Why Every Audit Report Carries a Caveat

1 You cannot audit everything

An audit examines a SAMPLE of evidence within a finite time. That is not a weakness — it is the definition of auditing.

2 Judgement-based sampling

Chosen by risk: the newest process, the biggest aspect, the area with previous findings, the shift nobody watches. Most internal audits use this.

3 Statistical sampling

Random selection allowing a confidence level to be stated. Used where populations are large and homogeneous, e.g. thousands of waste tickets.

4 State the basis and its limit

Record what you sampled and why — and put the sampling statement in the report. Absence of a finding is NOT proof of conformity.

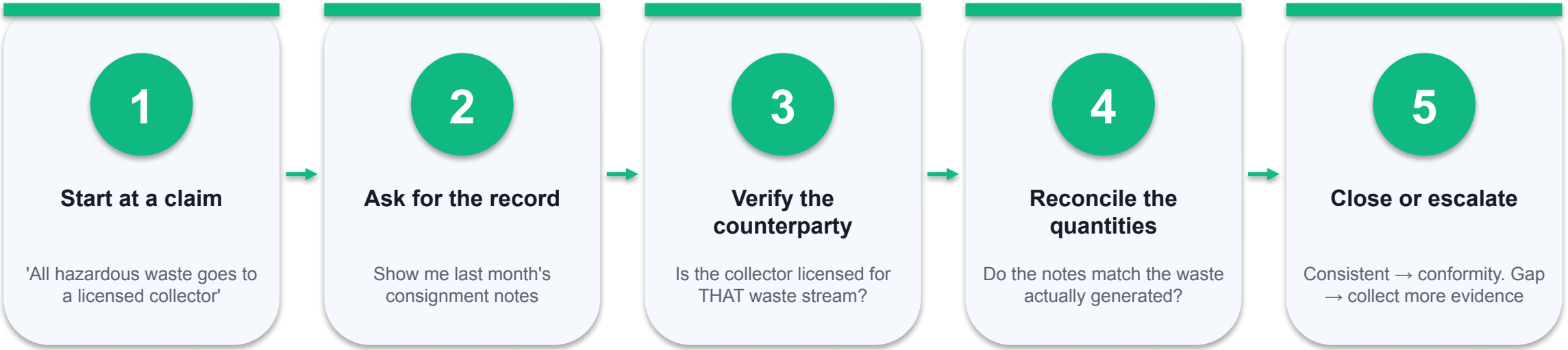
THE POINT

If you cannot explain WHY you looked at those three drums and not the other forty, your finding is vulnerable at the closing meeting.

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TRACEABILITY IN PRACTICE

Following an Audit Trail



THE POINT

An audit trail is a chain: claim → record → verification → reconciliation. Each link is a chance to find the gap, and the whole chain is what makes a finding defensible six months later.

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ISO 19011 §6.3

Stage 2 — Preparing: What Actually Happens

1. **Review documented information**
   The EMS documentation, previous audit reports, the aspects register, the legal register, licences and monitoring data. Determine adequacy BEFORE you go on site.

2. **Prepare the audit plan**
   Objectives, criteria, scope, locations, dates and times, methods, roles, and the opening and closing meeting times. Agreed with the auditee.

3. **Assign work to the audit team**
   Who audits which process, which clause, which shift, which site.

4. **Prepare work documents**
   The checklist, sampling plans and forms for recording evidence. These are working papers — they are retained.

5. **Decide the sampling approach**
   How many, which ones, over what period — and be ready to justify it. Sampling is why every audit report carries the uncertainty caveat.

An audit that is under-prepared cannot be rescued on site. Most weak audits fail here.

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The Audit Checklist — What Makes One Good

1 Organisation-specific

Built from THIS auditee's processes, aspects and obligations. A generic clause-by-clause tick-list is the single most common weakness.

2 Risk-weighted

More lines against the significant aspects, the recent changes and the previous findings; fewer against low-risk administrative processes.

3 Three columns minimum

The CLAUSE or requirement · the QUESTION you will ask · the EVIDENCE you expect to see. The third column is what most checklists omit.

4 A guide, not a script

Follow the audit trail where it leads. A checklist that stops you investigating something you have just discovered is being used wrongly.

THE POINT

Bureau Veritas and TÜV SÜD both single out checklist customisation as a core skill. Build it from the aspects register, not from the standard's contents page.

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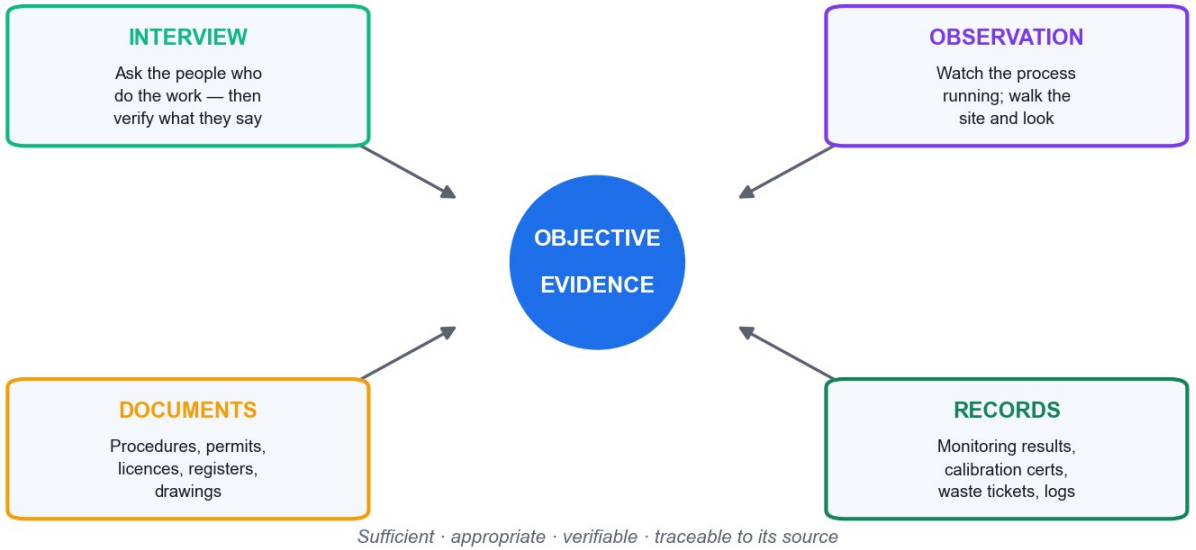
COPY THIS STRUCTURE

A Risk-Based Checklist Line — the Shape

Clause / requirement	The question I will ask	Evidence I expect to see
6.1.2 aspects · change	How were aspects re-assessed after the new line was commissioned?	Dated register entry for the new process; the change record
6.1.3 obligations	Which discharge limits apply to you, and how did you determine that?	Licence/Written Approval; legal register with applicability noted
9.1.1 monitoring	How do you know the monitoring result is valid?	Calibration certificate in date; accredited lab report
9.1.2 compliance	How often do you evaluate compliance, and what did the last one find?	Compliance evaluation record and the actions arising
10.2 corrective action	How did you verify last year's corrective action worked?	Effectiveness review record — NOT just a 'closed' status

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Audit Evidence — the Four Sources



Interview
What people say. The weakest form on its own — always verify it.

Observation
What you see happening. Strong, but a snapshot in time.

Documents
Procedures, permits, licences, drawings — what is SUPPOSED to happen.

Records
Monitoring results, certificates, logs — evidence it DID happen.

The test
Sufficient, appropriate, verifiable and TRACEABLE to its source.

An interview answer alone rarely closes a line of enquiry. Either you saw it, or there is a record of it, or you have not verified it.

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WHERE MOST NEW AUDITORS FAIL

Questioning Technique — the Funnel



Roughly 70% open and probing, 30% closed. A funnel that STARTS closed ('do you have a procedure?') gets a yes and teaches you nothing.

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THE SYSTEMATIC QUESTION SET

5W1H — Guaranteeing You Miss No Dimension

	The question	Applied to an EMS process
What	What is done? What is the requirement?	What environmental aspects arise from this activity?
Why	Why is it done this way? Why this control?	Why was this aspect rated not significant?
Who	Who does it? Who is competent or authorised?	Who is authorised to sign the waste consignment note?
Where	Where is it done? Where is it recorded?	Where is the spill kit, and where is the nearest drain?
When	When is it done? How often? When last?	When was the emergency response last tested?
How	How is it controlled, measured, verified?	How do you verify the disposal contractor is licensed?

Two extensions worth teaching: 'how much / how many' quantifies, and 'what if' tests the abnormal and emergency conditions clause 6.1.2 and 8.2 require.

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Interviewing — What Works and What Destroys Evidence

✗ NEVER DO THIS

- Leading: 'you do calibrate that, don't you?'
- Compound: three questions in one breath
- Interrogating: rapid-fire closed questions
- Talking more than the auditee
- Consulting: 'what you should do is…'
- Naming individuals in a finding
- Interviewing only managers
- Accepting 'we always do it that way'

✓ DO THIS INSTEAD

- Open first: 'walk me through what happens…'
- One question, then WAIT — the pause does the work
- Explain why you are there and why you take notes
- Target: the auditor talks less than 30% of the time
- Note the gap; let the auditee design the fix
- Address the system, never the person
- Managers describe the procedure; operators reveal the practice
- 'Show me' — convert every claim into evidence

Interviewing is the largest published gap in the commercial market — only two of seven major providers teach questioning technique at all. It is also where new internal auditors most often fail.

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A NOTE-TAKING DISCIPLINE — NOT AN ISO TERM

PLOD — Capturing Traceable Evidence as You Find It

1

P — Person

The ROLE interviewed, never the individual's name: 'the Warehouse Supervisor'.

2

L — Location

Building, area, line, tank: 'Chemical Store B, external yard'.

3

O — Object

The physical thing with its identifier: 'drums WD-104/105/106'.

4

D — Document

Document identity AND revision or date: 'EMS-P-08 Rev 4, cl. 5.3'.

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THE TEST OF A GOOD NOTE

If your note has none of P, L, O or D, you have an impression — not evidence.

A finding must be re-findable. Another auditor, six months later, should be able to go to the same place, ask for the same document and reach the same conclusion.

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ISO 19011 §6.4.2

The Opening Meeting — the Standard Agenda

1. **Introductions and roles**
   Audit team, auditee representatives, guides and observers.
2. **Confirm the audit plan**
   Objectives, scope, criteria, timetable and the closing meeting time.
3. **Confirm communication channels**
   Who the audit team reports to during the audit; how issues are escalated.
4. **Explain the method and the SAMPLING basis**
   And that sampling means the audit carries an element of uncertainty.
5. **Explain how findings will be graded and reported**
   Major, minor, OFI — and what each means here.
6. **Confirm safety, escort and facility arrangements**
   PPE, restricted areas, emergency procedures.
7. **Invite questions**
   Confirm the auditee understands and agrees before fieldwork begins.

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ISO 19011 §6.4.10

The Closing Meeting — the Standard Agenda

1. **Thank the auditee**
   And restate the audit objective and scope.
2. **Restate the sampling caveat**
   Findings are based on sampled evidence — absence of a finding is not proof of conformity.
3. **Present the findings**
   Each with its evidence and its grade, so they are understood and acknowledged.
4. **Present the audit conclusion**
   On conformity and on effective implementation and maintenance.
5. **Agree the corrective action process**
   Owners, timeframes and how effectiveness will be verified.
6. **Record any diverging opinions**
   Unresolved disagreements are REPORTED, not suppressed — that is fair presentation.
7. **Confirm the report and follow-up**
   Distribution, timing and the follow-up arrangement.

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Engaging the Stakeholders — They Are Not All the Same

1

Operations staff

Your primary evidence source. Interview at their normal workplace, put them at ease, never blame.

2

Process owners / managers

Describe the intended procedure and own the corrective actions. Contrast what they say with what operators do.

3

Top management

Opening and closing meetings; leadership (5.1), resources (7.1) and the programme-level findings.

4

Client representatives

NOT the auditee. Gatekeepers for site access — get permission in advance and protect confidentiality.

5

Regulators (NEA, PUB)

Engaged INDIRECTLY — through licences, submitted returns and correspondence. An internal auditor does not interview them.

6

Contractors on site

Work 'under the control of' the organisation — clause 7.2 competence and 8.1 outsourced control apply to them.

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ACTIVITY 2 · 60 MIN

Activity 2 — AquaPure — Planning the Engagement and the…

AquaPure's workload has tripled on a new government contract, its audit procedure is five years old, and two 'closed' nonconformities have no verification evidence. Plan the engagement stage by stage, build the checklist, and decide how you will handle each stakeholder group.

1. Walk the ISO 19011 §6.2–6.7 lifecycle and list actions for AquaPure at each stage
2. Draft the audit plan: objective, criteria, scope, sampling and timetable
3. Write 8 checklist lines with the clause, the question and the evidence sought
4. Build the stakeholder engagement matrix for the four groups
5. Rehearse the opening meeting and one interview using the questioning funnel

YOU'LL PRODUCE An Audit Plan + an 8-line risk-based Audit Checklist + a Stakeholder Engagement Matrix for the AquaPure wastewater audit, plus a rehearsed opening-meeting script.

Materials: Activity 2 brief · ISO 19011:2018 lifecycle map · Checklist and plan templates · Full step-by-step guide: Learner Guide, Activity 2

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TOPIC 02 · SELF-CHECK

Activity 2 — Check Your Work

YOU ARE DONE WHEN

Your checklist has at least 8 lines, every line names a clause AND the specific evidence you will ask to see, at least 4 lines target mobile operations / calibration / training / the previous NCs, and your matrix treats client representatives differently from the auditee.

IF YOU ARE STUCK — WHAT USUALLY GOES WRONG

- Your checklist is a generic clause-by-clause tick-list. Rebuild it from AquaPure's own risks: mobile units, calibration, training, discharge monitoring.
- You missed the third checklist column. Every line needs the EVIDENCE you expect to see, not just the question.
- You classified the 5-year-old procedure as an audit finding. It is a PROGRAMME finding against ISO 14001 cl. 9.2.2.
- You accepted 'closed' on the two previous NCs. Clause 10.2(d) requires a review of EFFECTIVENESS — no evidence means no closure.
- You planned to interview the regulator. Internal auditors engage NEA indirectly, through licences, returns and correspondence.
- You forgot client-site access. If you cannot reach the mobile units, the audit is not feasible as scoped — raise it at initiation, not in the report.

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Lunch Break

45 minutes

TOPIC 03

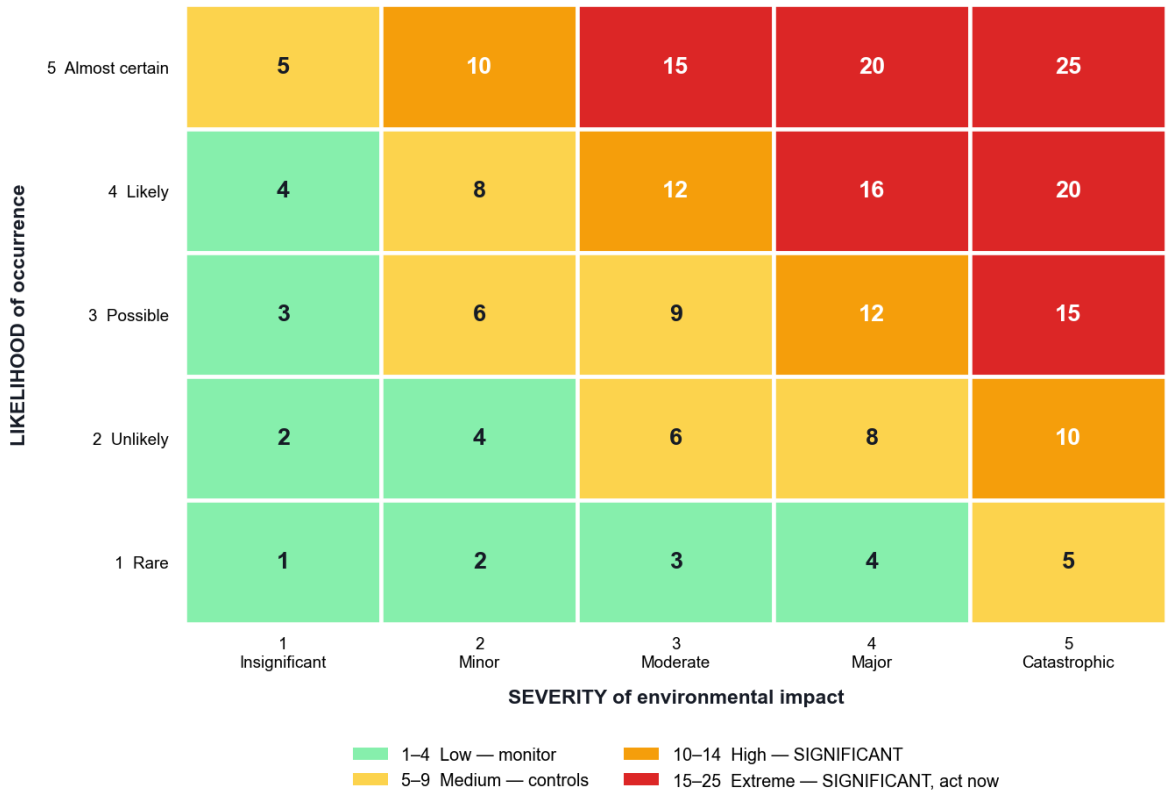
Risk Assessment

Aspect/impact significance · risk matrices and FMEA · nonconformity grading · root cause analysis · CARs · the audit report

03

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Evaluating Significance — the 5×5 Matrix



Score = Severity × Likelihood

The most common method, and defensible if the criteria are documented.

Threshold ≥ 10 = significant

High and Extreme bands. The threshold itself is what an auditor challenges.

ISO prescribes NO method

6.1.2 requires only that significance is determined BY ESTABLISHED CRITERIA.

The real audit question

Not 'is your method right?' but 'is it documented, criteria-based and consistently applied?'

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WHY SCORING ALONE IS NOT ENOUGH

The Override Rules — What Makes an Aspect Automatically Significant

1 A legal limit applies

The aspect is subject to a regulatory limit, licence or permit condition — regardless of its arithmetic score.

2 A regulator has acted

A notice, warning, fine or prosecution in the last three years.

3 A substantiated complaint

An interested party has complained and the complaint was upheld.

4 Controlled substances or waste

The aspect involves a listed hazardous substance or toxic industrial waste.

THE POINT

Without override rules, a licensed discharge can score 'medium' and drop off the significant list — the classic finding where the aspects register has become disconnected from the legal register.

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Three Significance Methods — Pick One and Document It

Method	How it scores	Best for
Severity × Likelihood	5×5, score 1–25; significant at ≥ 10	SMEs and first registers — simple and defensible
Multi-criteria weighted	Severity × Frequency × Legal × Stakeholder, or a weighted sum	Larger organisations — builds in obligations and interested parties
FMEA (S × O × D)	RPN 1–1000; act at ≥ 100; any S ≥ 9 acts regardless	Abnormal and emergency aspects where DETECTION matters

FMEA adds the dimension the others miss: can we DETECT the failure before it becomes an impact? The cheapest risk reduction is almost always better detection — which is exactly what clause 9.1 monitoring is for.

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TAKE THESE ON SITE

Auditing the Significance Method — Questions That Find Weakness

1. **Show me the documented criteria.**
   If the criteria live only in a spreadsheet formula or someone's head, that is a 6.1.2 finding.

2. **Who approved them, and when?**
   No approval, no date, no version control.

3. **Would two assessors score this the same?**
   Test it live — re-score one aspect with the process owner.

4. **This has a legal limit but scored 6. Explain.**
   Tests whether override rules exist.

5. **Show me how each significant aspect reaches a control.**
   6.1.2 → 8.1 → 9.1. A register nothing references is shelf-ware.

6. **Why did significant aspects drop from 30 to 6?**
   Tests whether the threshold was quietly moved to reduce work.

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WHEN SEVERITY × LIKELIHOOD IS TOO COARSE

FMEA for Environmental Risk — S × O × D

Dimension	What it rates	The audit insight
Severity (S)	How bad the environmental consequence is, 1–10	You cannot usually reduce it — a solvent stays toxic
Occurrence (O)	How likely the failure mode is, 1–10	Reduced by maintenance, design and competence
Detection (D)	Ability to detect BEFORE impact — INVERTED scale, 1 = certain	The cheapest and fastest RPN reduction available
RPN = S × O × D	1–1000; act at ≥ 100; any S ≥ 9 acts regardless of RPN	Links directly to clause 9.1 monitoring and measurement

Worked example: cracked bund, S=9 O=4 D=7 → RPN 252. Add a leak-detection sensor and D falls to 2 → RPN 72. Same hazard, same likelihood, but now you find out before it reaches the soil.

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CLAUSE 6.2.2 · A RELIABLE CHECKLIST

Environmental Objectives — the 6.2.2 Five-Part Test

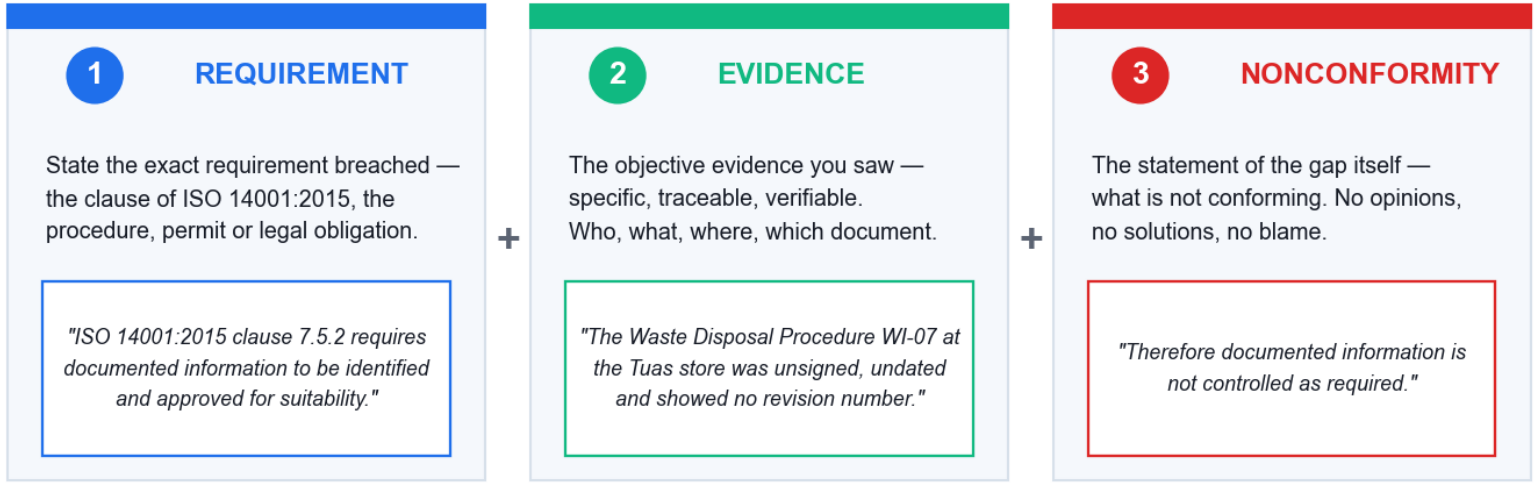
Requirement	The audit question	Weak answer
What will be done	What are the actual actions and programme?	'Improve energy efficiency' — that is the objective, not the plan
What resources	Is the budget approved and released?	No capex, no headcount — a paper objective
Who is responsible	Which NAMED role owns it?	'The EHS department'
When completed	Target date and milestones?	'Ongoing'
How evaluated	Which indicator, measured how, how often, by whom?	No indicator, or an indicator with no calibrated measurement

Trace every objective BACKWARDS to a significant aspect, obligation or risk, and FORWARDS to 9.1 monitoring and 9.3 management review. An orphan objective means the 6.1 → 6.2 link is broken.

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THE MOST IMPORTANT SKILL IN THIS COURSE

Writing a Nonconformity — the Three-Part Structure



THE RULE

A nonconformity statement without all three parts is an opinion. Requirement + Evidence + Gap — every time.

Requirement + Evidence + the statement of the gap. Miss any one and you have written an opinion, which the auditee is entitled to reject.

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AND WHERE THE CRITERIA COME FROM

Grading the Finding — Major, Minor or OFI

MAJOR	MINOR	OFI
• Total absence of a required process	• A single lapse against a requirement	• NOT a nonconformity
• A systemic breakdown of the EMS	• An isolated slip in an otherwise working process	• The requirement IS met
• A legal / permit breach	• Does not break the system	• A suggestion to do better
• Several minors against one clause	• Still needs correction + CA	• Never forced on the auditee
• Puts certification at risk		• Recorded separately

Grade the finding against the REQUIREMENT and the EVIDENCE — never against how the auditee reacts.

ISO 19011 says only that findings 'can be graded'. The major/minor criteria come from ISO/IEC 17021-1 and from your own organisation's audit procedure — so define them in the procedure and apply them consistently.

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REWRITE EVERY WEAK ONE

Nonconformity Statements — Bad and Good

✗ WEAK — will be rejected

- 'Housekeeping in the chemical store is poor.'
- → opinion, no requirement, no evidence
- 'The operator did not follow the procedure.'
- → blames a person, names no requirement
- 'You should install a new bund.'
- → that is consulting, not a finding
- 'Training records were not very good.'
- → vague, not traceable, not re-findable

✓ STRONG — evidence-based

- 'ISO 14001:2015 cl. 8.1 requires operational controls...
- ...On 14 Aug 2026, three 200 L drums (WD-104/105/106) in Chemical Store B bore no contents or hazard labels...
- ...therefore operational control for hazardous waste storage is not implemented as required.'
- Requirement + Evidence + Gap
- Traceable: another auditor can re-find it
- Addresses the SYSTEM, not the person

Write findings so that a reader who was not there understands exactly what was seen, where, and which requirement it breaches.

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STOP AT A PROCESS, NOT AT A PERSON

Root Cause Analysis — 5 Whys Worked Through



If your 5 Whys ends at 'the operator forgot', you have found a symptom. Push until you reach a step that is missing from a process — that is what corrective action can actually fix.

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CLAUSE 10.2 VS 10.3

Correction vs Corrective Action vs Improvement

CORRECTION Fixes the SYMPTOM	CORRECTIVE ACTION Eliminates the CAUSE	CONTINUAL IMPROVEMENT Raises PERFORMANCE
• Clean up the spill	• Change the procedure	• Redesign to remove the hazard
• Re-label the drums	• Add the calibration to the schedule	• Substitute a less harmful input
• Re-issue the missing record	• Re-train and verify competence	• Improve the EMS itself
• Immediate, necessary — but the problem WILL come back	• Prevents RECURRENCE — this is what clause 10.2 requires	• Goes beyond conformity — clause 10.3

Clause 10.2 requires BOTH: react and correct, AND evaluate the need to eliminate the cause so it does not recur. A CAR that only lists corrections is an incomplete CAR.

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Writing an Effective Corrective Action Request

1. **The nonconformity statement**
   Requirement + evidence + gap, with the clause cited.
2. **The grade and its justification**
   Major, minor or OFI — and WHY, against your defined criteria.
3. **Root cause**
   Reached by 5 Whys or fishbone; must be a process or system cause.
4. **Correction**
   What fixes the immediate problem and contains the consequences — including mitigating the environmental impact (10.2a).
5. **Corrective action**
   What eliminates the cause so it does not recur OR OCCUR ELSEWHERE — and whether similar NCs exist in other areas (10.2b).
6. **Owner and due date**
   A named role and a real date.
7. **Effectiveness verification**
   How and when you will confirm it worked — required by 10.2(d) and retained under 10.2.

'Re-trained the operator' is the most over-used and least effective corrective action in industry. Ask: if that person leaves tomorrow, does the problem return?

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ISO 19011 §6.5 · A3

The Audit Report — What Must Be In It

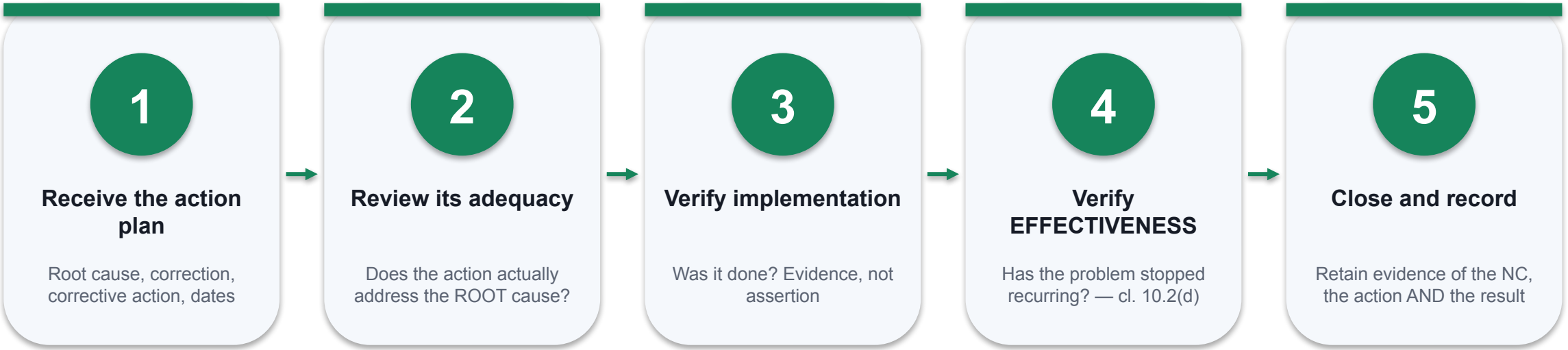
1. **Audit objectives, scope and criteria**
   Including what was EXCLUDED.
2. **Audit team, auditees and dates**
   Who audited, who was audited, when.
3. **The audit findings**
   Each with its evidence, its clause and its grade.
4. **The audit conclusion**
   On conformity AND on effective implementation and maintenance — clause 9.2.1 requires both.
5. **The sampling statement**
   That the audit was a sampling exercise and therefore carries uncertainty — the single most commonly OMITTED element.
6. **Opportunities for improvement**
   Kept clearly SEPARATE from nonconformities.
7. **Agreed actions, owners and dates**
   And the follow-up arrangement.
8. **Distribution list**
   Reported to relevant management — ISO 14001 cl. 9.2.2.

Report item (k) — the sampling caveat — protects both the auditor and the organisation. Absence of a finding is never proof of conformity.

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ISO 19011 §6.7 · ISO 14001 CL. 10.2

Stage 6 — Audit Follow-Up: the Stage Everyone Skips



THE POINT

'Closed' is a status, not a conclusion. Clause 10.2(d) requires a review of the EFFECTIVENESS of corrective action, and 10.2 requires the result to be retained. Closure without that evidence is itself a nonconformity.

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THE TEST: WOULD THIS SURVIVE THE PERSON LEAVING?

Corrective Actions — Weak and Strong

✗ WEAK — the problem will return

- 'Operator re-trained.'
- → if that person leaves, the gap returns
- 'Reminded staff at the toolbox talk.'
- → no change to any process
- 'Increased management awareness.'
- → unmeasurable, unverifiable
- 'Will monitor more closely.'
- → no owner, no frequency, no criterion

✓ STRONG — the cause is removed

- 'Amended MOC procedure PR-12 to require an
- environmental re-assessment before any volume
- or process change — effective 1 Sep 2026.'
- 'Added bund integrity testing to the PM schedule
- at 12-month frequency, owner: Facilities Manager.'
- 'Installed continuous pH monitoring with an
- interlocked shut-off on the discharge line.'
- Process changed · owner named · date set · verifiable

Re-training is the most over-used corrective action in industry. It is sometimes right — but only when the root cause genuinely IS a competence gap, and then clause 7.2 requires you to evaluate the effectiveness of the training.

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THE PITFALLS

Common Auditor Mistakes — Recognise Them In Yourself

1. **Auditing the person, not the process**
   'Why didn't YOU fill this in?' kills cooperation and finds nothing.

2. **Raising an NC against a 'should'**
   Only 'shall' statements are requirements.

3. **Demanding documents that are not required**
   No procedure, no manual, no preventive action procedure is required.

4. **Consulting mid-audit**
   You become the author of what you must audit next year.

5. **Accepting an assertion as evidence**
   'We always do that.' Show me.

6. **Stopping the 5 Whys at human error**
   Human error is a symptom. Find the process gap.

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ACTIVITY 3 · 75 MIN

Activity 3 — EcoTech Electronics — Risk Assessment, CARs…

EcoTech failed an NEA inspection for exceeding its copper discharge limit, three hazardous waste containers are unlabelled, the spill bund is cracked, and the EHS Manager post has been vacant for three months. Assess the risks, write the CARs, and draft the report.

1. Build the aspect/impact register for the five problem areas
2. Score each risk on the 5×5 matrix and set the significance threshold
3. Run 5 Whys on the copper exceedance to reach the true root cause
4. Write 3 CARs — requirement + evidence + nonconformity, graded
5. Draft the audit report sections and the conclusion on effectiveness

YOU'LL PRODUCE A prioritised environmental risk assessment, three fully-written Corrective Action Requests with root cause analysis, and the skeleton of the audit report to EcoTech's management.

Materials: Activity 3 brief · 5×5 risk matrix · CAR form · Audit report template · Full step-by-step guide: Learner Guide, Activity 3

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Activity 3 — Check Your Work

YOU ARE DONE WHEN

Your register separates aspect from impact correctly on every line, covers normal / abnormal / emergency conditions, applies the override rule so the copper exceedance is significant, your 5 Whys reaches a process cause (not a person), and each of your three CARs contains requirement + evidence + nonconformity with a stated grade.

IF YOU ARE STUCK — WHAT USUALLY GOES WRONG

- You wrote 'copper' or 'pollution' in the aspect column. The aspect is the INTERACTION ('discharge of trade effluent containing copper').
- You scored the copper exceedance as medium and left it off the significant list. A breached legal limit is significant regardless of the arithmetic.
- Your 5 Whys stopped at 'the operator did not check'. Push to the missing process step — that is what corrective action can fix.
- Your CAR lists only corrections (re-label, repair). Clause 10.2 also requires action to eliminate the CAUSE.
- You graded everything Major. Grade against your defined criteria and justify each one; over-grading destroys credibility as fast as under-grading.
- You omitted the sampling statement from the report. It is the most commonly missed element and it protects you.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES DELIVERED

What You Achieved

1

The framework (LO1)

The IPPF, the Code of Ethics, ISO 14001:2015 clause by clause, and how context, aspects and obligations drive the EMS.

2

The audit process (LO2)

ISO 19011's seven principles, the programme/audit distinction, the six-stage lifecycle, checklists, evidence and interviewing.

3

Risk and reporting (LO3)

Significance evaluation, risk matrices and FMEA, root cause analysis, corrective action requests and the audit report.

4

Three real engagements

GreenPack scoping, AquaPure planning, EcoTech risk and CARs — the same shape as the assessment.

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WHAT TO DO ON MONDAY

Your First Real Internal Audit — a Checklist

1. **Confirm your independence**
   You cannot audit your own area. Declare any prior involvement.

2. **Read before you walk**
   The aspects register, the legal register, previous reports and open corrective actions.

3. **Build the checklist from THEIR risks**
   Not from the standard's contents page.

4. **Verify everything you are told**
   Interview, then observe, then check the record.

5. **Write findings that are re-findable**
   Requirement + evidence + gap, with P, L, O, D.

6. **Follow up — always**
   10.2(d) requires you to review whether the action was effective.

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CONTINUE THE JOURNEY

Recommended Follow-On Courses

1. **ISO 9001:2015 Internal Auditor Training**
   WSQ funded · Tertiary Infotech Academy

2. **ISO 45001:2018 Internal Auditor Training**
   WSQ funded · Tertiary Infotech Academy

3. **Sustainability Reporting and ESG Essentials**
   WSQ funded · Tertiary Infotech Academy

4. **Carbon Footprint Accounting and Management**
   WSQ funded · Tertiary Infotech Academy

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AFTER THE COURSE

Support — We're Here to Help

1 Email

enquiry@tertiaryinfotech.com
For course, certificate and funding enquiries during and after the class.

2 Telephone

+65 6100 0613
Office hours, Monday to Friday.

3 Website

www.tertiarycourses.com.sg
Course schedules, funding information and the full WSQ catalogue.

THE POINT

Your certificate is issued after you are assessed Competent, meet the 75% attendance requirement and complete the TRAQOM survey on the LMS.

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FINAL ASSESSMENT

Assessment

1. **Written Assessment (SAQ)**
   4 short-answer questions · 60 minutes · K1–K4.

2. **Case Study (CS)**
   3 scenario-based questions · 60 minutes · A1–A3.

3. **Open book**
   Slides, Learner Guide, activity briefs and your own notes.

4. **Attendance first**
   Take the Assessment digital attendance before you start.

5. **Submit on the LMS**
   https://lms-tms.tertiaryinfotech.com

6. **Sign the record**
   Sign the Assessment Summary Record before you leave.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (SAQ) 1 hr, then the Case Study 1 hr — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

1. Trainer displays the SSG digital attendance QR code
2. Scan the QR with your phone camera
3. Submit your attendance — AM, PM and Assessment
4. Repeat on BOTH days of the course
5. 75% minimum attendance for assessment and funding

Digital attendance is mandatory for WSQ-funded courses and is the record SSG uses to determine your funding eligibility.

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SEE YOU AT THE OPENING MEETING

Thank You!

You can now plan, conduct, report and follow up an internal audit of an ISO 14001:2015 environmental management system — and write findings that stand up.

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